(allow 0.002/0.182 and allow 1 → 4 sig. fig.)

2 **(a)** **(i)** grav. pot. energy = GM_1M_2/R 1

energy $= \{6.67 \times 10^{-11} \times 197 \times 4 \times (1.66 \times 10^{-27})^2\}/9.6 \times 10^{-15}$ 1

$= 1.51 \times 10^{-47}$ J 1 **[3]**

(ii) elec. pot. energy = $Q_1Q_2/4\pi\varepsilon_0R$ 1

energy $= \{79 \times 2 \times (1.6 \times 10^{-19})^2\}/4\pi \times 8.85 \times 10^{-12} \times 9.6 \times 10^{-15}$ 1

$= 3.79 \times 10^{-12}$ J 1 **[3]**

*(For the substitution, -1 each error or omission to max 2 in **(i)** and in **(ii)**)*

(b) electric potential <u>energy</u> >> gravitational potential <u>energy</u> 1 **[1]**

(c) *either 6* MeV = 9.6 *x* 10^{-13} J or 3.79 *x* 10^{-12} J = 24 MeV 1

not enough energy to get close to the nucleus 1 **[2]**